Comparative Analysis of Generative Models for Artistic Image Synthesis on ArtBench-10

Arthur Sophiatti[2022115599] and Faedo Giovanni[2025267503]

University of Coimbra, Coimbra, PT

Abstract. This paper presents a comparative study of three major generative modeling paradigms applied to the ArtBench-10 dataset: Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Diffusion Models. The objective is to generate artistic images at a resolution of 32×32 across ten stylistic categories.
Model development was first conducted on a stratified 20% subset of the training set to enable efficient hyperparameter tuning, followed by final training on the full dataset using the best configurations. Performance was evaluated both qualitatively and quantitatively using Fréchet Inception Distance (FID) and Kernel Inception Distance (KID), with results averaged across multiple random seeds.
The experiments reveal clear trade-offs between the evaluated approaches. VAEs provide smooth and structured latent representations, but produce lower visual fidelity. DCGAN achieved the best quantitative performance, generating sharper and more realistic samples. Diffusion models produced competitive results, although limited by the low image resolution and computational budget.
These findings highlight how model choice depends on the balance between sample quality, latent controllability, and training efficiency.

Keywords: Generative Artificial Intelligence · ArtBench-10 · Variational Autoencoders · Generative Adversarial Networks · Diffusion Models · Image Synthesis

1 Introduction

Generative modeling has become a central topic in modern Artificial Intelligence. Unlike discriminative methods, which learn decision boundaries or labels, generative models aim to approximate the underlying data distribution in order to synthesize new samples. Recent advances in deep learning have significantly improved the quality of generated images, making generative methods relevant for creative applications, simulation, and representation learning.
Among the most influential approaches are Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Diffusion Models. These families rely on different learning principles and present complementary strengths. VAEs offer stable optimization and structured latent spaces, GANs are known for producing sharp and realistic images, while Diffusion Models have recently achieved strong results through iterative denoising procedures.

In this project, we study generative modeling using the *ArtBench-10* dataset [1], which contains artworks grouped into ten artistic styles. Compared with natural-image datasets, artwork generation is particularly challenging because models must capture not only shapes and colors, but also style-dependent characteristics such as texture, composition, contrast, and abstraction. In addition, the low image resolution (32×32) limits the amount of available visual information.

The objective of this work is to implement, train, and compare representative models from the three paradigms above under a controlled experimental protocol. Development is first performed on a provided 20% subset for efficient experimentation, followed by final training on the full dataset. Model quality is assessed through visual inspection and through Fréchet Inception Distance (FID) and Kernel Inception Distance (KID), allowing a consistent comparison of realism and diversity.

2 Dataset and Problem Formulation

The objective of this project is to train generative models capable of synthesizing artwork images that are both visually plausible and stylistically diverse. Unlike natural-image datasets such as CIFAR-10, ArtBench-10 contains artistic images with higher stylistic variability and less rigid semantic structure.

2.1 The ArtBench-10 Dataset

ArtBench-10 [1] contains 60,000 RGB images divided into 50,000 training samples and 10,000 test samples across ten artistic styles: *Art Nouveau, Baroque, Expressionism, Impressionism, Post-Impressionism, Realism, Renaissance, Romanticism, Surrealism,* and *Ukiyo-e.*
Our preliminary analysis of the dataset revealed that it is **perfectly balanced**, with each style containing exactly 5,000 training samples. Due to this uniform distribution, no additional oversampling or data augmentation techniques were required to address class imbalance, ensuring a stable and unbiased learning process across all artistic categories.

2.2 Data Strategy and Pre-processing

To ensure efficient development, we follow a two-stage approach as specified in the project guidelines:

1. **Exploratory Phase**: Models are trained on a 20% stratified subset of the training data. This allows for faster hyperparameter tuning and architecture comparison.
2. **Final Training**: The best-performing model is retrained on the full training dataset to achieve maximum generative quality.

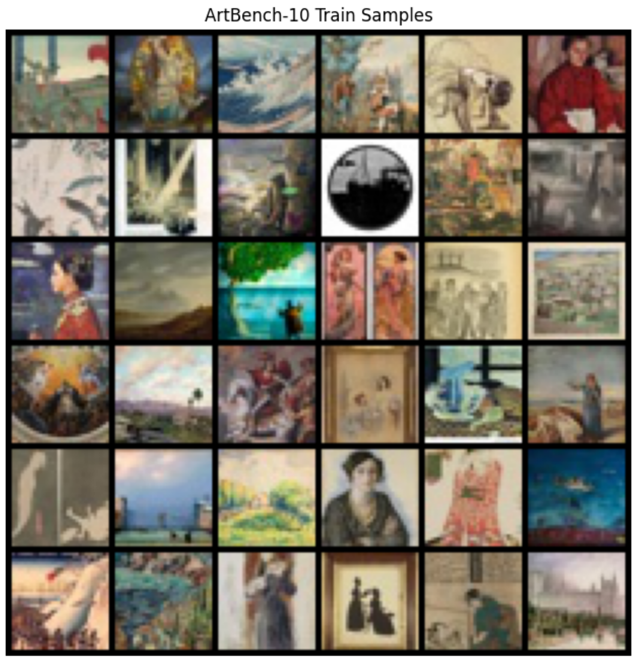

Fig. 1. Representative samples from ArtBench-10 at 32×32 resolution, illustrating the diversity of artistic styles used in this project.

In this work, we utilize the standardized 32×32 RGB version of the dataset. This lower resolution, while reducing the computational overhead, requires the generative models to capture the essence of each style within a very limited spatial budget ($32 \times 32 \times 3 = 3,072$ dimensions).
The **preprocessing** pipeline was adapted to each model family. VAE-based models were trained on images normalized to $[0, 1]$, while GAN and diffusion models used $[-1, 1]$ normalization to match output activations and noise parameterization.

3 Methodology

This section summarizes the generative architectures evaluated in this study. We progressively considered reconstruction-based, adversarial, and diffusion-based approaches.

3.1 Autoencoder Baselines

Autoencoders were first used as reconstruction baselines. Given an input image x, an encoder maps it into a latent representation z, which is then decoded to reconstruct the original sample.

Dense Autoencoder The Dense Autoencoder processes flattened inputs through fully connected layers. While computationally simple, this architecture ignores spatial locality and therefore struggles to reconstruct fine artistic textures.

Convolutional Autoencoder (ConvAE) The ConvAE replaces dense layers with convolutional blocks and downsampling operations. By exploiting spatial structure, it provides substantially better reconstructions and was used as the main deterministic baseline. The model is trained using a pixel-wise reconstruction loss defined as the Mean Squared Error (MSE) between the input image x and its reconstruction $\hat{x}$:

$$\mathcal{L}_{rec} = \|x - \hat{x}\|_2^2 \tag{1}$$

This objective corresponds to the **Mean Squared Error (MSE)** loss, which encourages faithful reconstruction of input images by minimizing the average squared difference between corresponding pixels. However, it does not impose any constraint on the latent distribution.

3.2 Variational Autoencoder

To enable stochastic generation, we extend the convolutional architecture into a Variational Autoencoder (VAE) [3]. Instead of mapping each sample to a single latent vector, the encoder predicts the parameters of a Gaussian distribution (μ, σ).
Latent samples are obtained through the **reparameterization** trick:

$$z = \mu + \sigma\epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{2}$$

The model is optimized using a combined objective composed of a reconstruction loss and a Kullback-Leibler (KL) divergence term:

$$\mathcal{L}_{VAE} = \mathcal{L}_{rec} + \beta\,\mathcal{L}_{KL} \tag{3}$$

where the KL divergence follow these equation:

$$\mathcal{L}_{KL} = -\frac{1}{2}\sum\left(1 + \log\sigma^2 - \mu^2 - \sigma^2\right) \tag{4}$$

The reconstruction loss is implemented as Mean Squared Error (MSE), while the KL divergence term encourages the learned latent distribution to remain close to a standard normal distribution $\mathcal{N}(0, I)$.
This objective encourages the model to learn a smooth and continuous latent space, making it suitable for both sampling and interpolation.

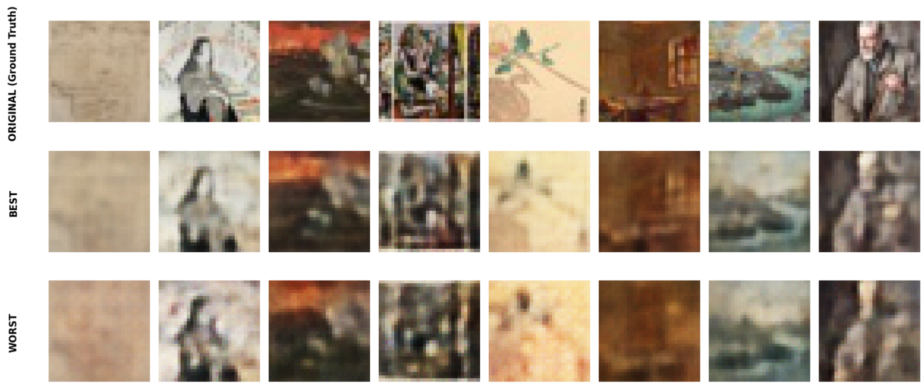

Fig. 2. Visual comparison of ConvVAE reconstruction quality. Top row: ArtBench-10 original samples (Ground Truth). Middle row: Reconstructions from the best-performing configuration. Bottom row: Reconstructions from the worst-performing configuration, showing significant loss of detail and increased blurriness.

3.3 Generative Adversarial Networks

We implemented a Deep Convolutional GAN (DCGAN) [5], consisting of a generator G and a discriminator D trained in an adversarial framework.
The generator maps a latent noise vector $z \sim \mathcal{N}(0, I)$ into the image space using a sequence of transposed convolutional layers, batch normalization, and ReLU activations. The final layer uses a tanh activation to produce images in the normalized range. The discriminator is a convolutional neural network that progressively reduces spatial resolution while increasing feature depth, outputting a scalar probability of the input being real or generated.
The model is trained using a standard binary cross-entropy adversarial loss.
The GAN is formulated as a minimax game between the generator and the discriminator:

$$\min_G \max_D V(D, G) = E_{x \sim p_{data}}[\log D(x)] + E_{z \sim p(z)}[\log(1 - D(G(z)))] \tag{5}$$

The discriminator aims to correctly classify real and generated samples, while the generator learns to produce samples that maximize the discriminator's probability of classifying them as real. This adversarial setup encourages the generator to produce sharper images with more realistic high-frequency details compared to reconstruction-based approaches such as Autoencoders and VAEs.

3.4 Diffusion Models

We also evaluated Denoising Diffusion Models [6], where generation is performed by progressively reversing a noising process.

Pixel-Space Diffusion : In Pixel Diffusion, the model learns to iteratively denoise images directly in pixel space ($32 \times 32 \times 3$). A UNet architecture is trained to reverse a fixed forward noising process over T timesteps. Pixel-space diffusion models remain computationally expensive [8], although recent work suggests that direct pixel-space transformers can achieve strong performance without latent compression.

Latent Diffusion : To reduce the computational cost, we also implement Latent Diffusion, where the diffusion process is applied in a compressed latent space learned by a Variational Autoencoder. The UNet operates on latent tensors instead of raw pixels, and the final images are reconstructed through the VAE decoder. This increment the efficiency while preserving semantic structure.

Both diffusion models are trained to predict the noise added to an image at each timestep. The objective is defined as a simple mean squared error loss:

$$\mathcal{L}_{diff} = E_{x,\epsilon,t}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2\right] \tag{6}$$

where ϵ is the true noise, and $\epsilon_\theta(x_t, t)$ is the noise predicted by the model at timestep t.
Minimizing this loss allows the model to learn the reverse diffusion process, progressively transforming noise into realistic images.

4 Experimental setup

This section details the hardware and software environment, the specific architectural configurations explored during the grid search, and the rigorous evaluation protocol used to validate our outcomes.
All experiments were conducted using the PyTorch framework, accelerated by a CUDA-enabled GPU (Google Colab T4). To ensure the reproducibility of our results, including weight initialization and data shuffling, a global seed was fixed to 42.

4.1 Hyperparameter Optimization

We performed a systematic grid search over key architectural and optimization hyperparameters for all model families. The explored search spaces are summarized in Table 1.
The goal of this phase was to identify stable and well-performing configurations that balance training stability, computational efficiency, and generative quality. Overall, all models were evaluated under multiple configurations, and the best-performing ones were selected based on a combination of FID and KID scores. This ensured a fair comparison across different generative paradigms under consistent evaluation criteria.

Table 1. Hyperparameter search space used for grid search across all generative models (ConvAE, ConvVAE, DCGAN, and Diffusion)

Model	**Parameter**	**Values Tested**
ConvAE	Latent dim (z)	128, 256
	Base filters (f)	32, 64, 128
	Learning rate (lr)	0.001, 0.0005
	Epochs (e)	20, 30
ConvVAE	Latent dim (z)	128, 256
	Base filters (f)	32, 64
	Learning rate (lr)	0.001, 0.0005
	KL weight (β)	0.1, 0.5, 0.7, 1.0
	Epochs (e)	20
DCGAN	Latent dim (z)	100, 128
	Filter depth (ngf, ndf)	(64, 64), (128, 64), (32, 64)
	Learning rate (lr_G, lr_D)	$(2e^{-4}, 2e^{-4}), (1e^{-4}, 4e^{-4})$
	Epochs (e)	20, 30
Diffusion	Timesteps (T)	250, 500, 1000
	Base channels (ch)	64
	Learning rate (lr)	$1e^{-4}$
	Scheduler	Linear, Cosine
	Variant	Pixel / Latent

4.2 Selection of Optimal Configuration

Following the exploratory phase on the 20% subset, we identified the best-performing configuration for each model family by analyzing the correlation between FID and KID scores. These best models were subsequently retrained on the full 50,000-image dataset. Table 2 summarizes the hyperparameters selected for the final evaluation.

Table 2. Summary of the best-performing configurations and their respective FID/KID scores for each model family.

Model Family	**Best Configuration Details**	**FID**	**KID**
ConvAE*	$z = 256, lr = 0.001, f = 64, e = 30$	82.32	0.049
ConvVAE	$z = 128, lr = 0.001, f = 64, e = 20, \beta = 0.1$	181.43	0.155
DCGAN	$z = 100, lr_G = 1e^{-4}, lr_D = 4e^{-4}, f = 64, e = 30$	101.99	0.063
Diffusion	Pixel, $lr = 1e^{-4}, T = 500$, Linear scheduler, $ch = 64$	134.11	0.114

Note: ConvAE is excluded from final evaluation as it is a deterministic compression tool and not generative; lacking a latent prior, it cannot sample novel images from noise

5 Analysis of Results

This section analyzes the experimental results, focusing on differences in representation learning, generative quality, and architectural trade-offs across model families.

5.1 Reconstruction vs Generative Capability

A clear distinction emerges between reconstruction performance and generative capability. While autoencoder-based models (ConvAE and ConvVAE) achieve strong reconstruction quality, this does not directly translate into effective generative modeling.

ConvAE : The ConvAE minimizes a pixel-wise reconstruction objective, which encourages accurate reproduction of training samples but does not impose any structure on the latent space. As a result, although reconstructions are sharp, the latent distribution is not meaningful for sampling.

ConvVAE : The ConvVAE introduces a probabilistic constraint through KL-divergence regularization. This forces the latent space to approximate a Gaussian prior, improving continuity and sampling consistency. However, this regularization also introduces an inherent trade-off: the model sacrifices high-frequency detail in favor of smoother latent structure, leading to blurrier reconstructions and lower perceptual fidelity.

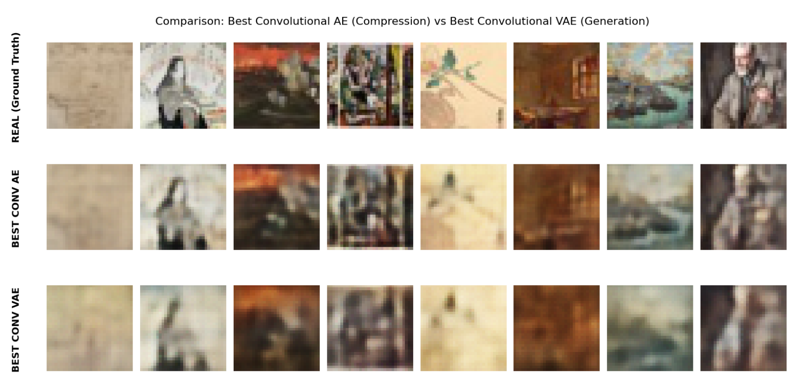

Fig. 3. Visual comparison of reconstruction quality between the best-performing ConvAE and ConvVAE configurations. The top row shows the ground truth (ArtBench-10 originals), while the other rows display the reconstructions. Note the smoother textures in the VAE output, a direct consequence of the KL-divergence regularization.

5.2 Latent Space Structure: Sampling and Interpolation

Beyond reconstruction performance, we analyze the structure of the learned latent space through random sampling and linear interpolation (latent morphing).

ConvAE : The latent space is optimized solely for reconstruction and is not constrained by any prior distribution. As a result, randomly sampled latent vectors do not correspond to meaningful points in the data manifold. This leads to unstable and visually incoherent samples, indicating that the latent space is highly irregular and poorly structured for generative purposes.

ConvVAE : The ConvVAE enforces a Gaussian prior over the latent space through KL-divergence regularization. This constraint significantly improves the global structure of the latent representation. Random sampling produces more coherent outputs,as sampled points are likely to lie within regions seen during training.
In ConvAE, transitions between two latent codes are often abrupt and visually inconsistent, reflecting the lack of continuity in the latent manifold. Conversely, ConvVAE exhibits smooth and semantically meaningful transitions, where changes in style, color distribution, and composition evolve gradually along the interpolation path.
These observations confirm that enforcing a probabilistic structure in latent space improves continuity and interpretability, even if it slightly degrades reconstruction fidelity. This trade-off is crucial for downstream generative tasks that require controllable sampling rather than deterministic reconstruction.

Fig. 4. Comparison of random sampling.

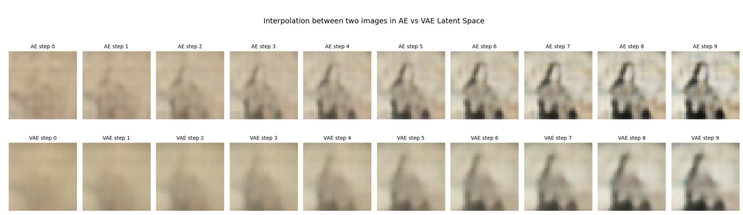

Fig. 5. Comparison of latent interpolation.

Fig. 6. Analysis of ConvAE vs ConvVAE latent space properties

5.3 Adversarial Training and Image Sharpness

DCGAN demonstrates significantly better performance in terms of visual quality and perceptual metrics. This follows from the adversarial training objective, which directly optimizes the generator to produce outputs indistinguishable from real images according to a learned discriminator.
Unlike reconstruction losses, adversarial loss does not penalize pixel-level deviations but instead focuses on distribution matching. This encourages the generation of sharper textures and higher-frequency details.
However, this advantage comes at a cost. Adversarial training is highly sensitive to hyperparameters and can suffer from instability or mode collapse. In our experiments, careful tuning of learning rates between generator and discriminator was required to achieve stable convergence.

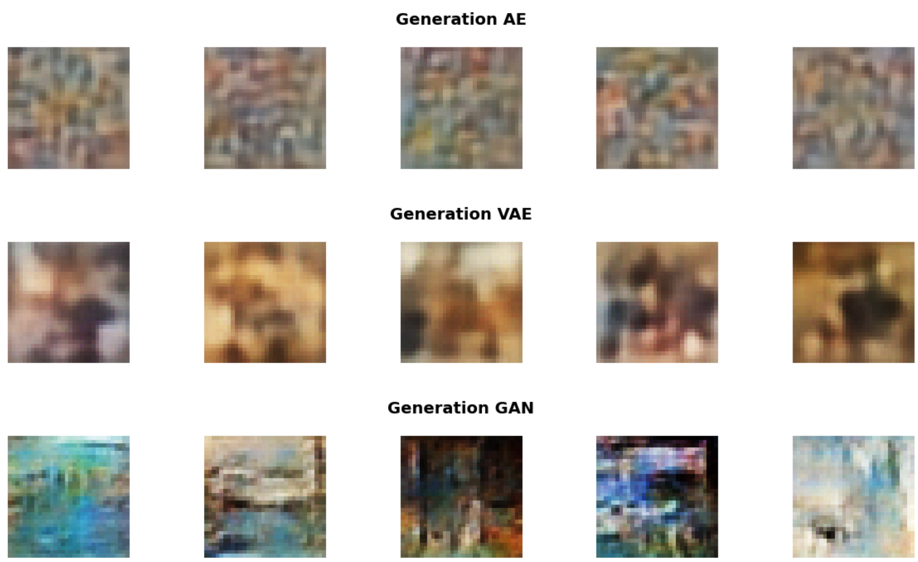

Fig. 7. Comparative visual samples of generated artworks from AE (top), VAE (middle), and GAN (bottom) models trained on ArtBench-10 at 32×32 resolution

5.4 Diffusion Models: Quality vs Efficiency Trade-off

Diffusion models provide a more stable alternative to adversarial training by modeling data generation as an iterative denoising process.
In our experiments, **Pixel Diffusion** achieves better visual fidelity than Latent Diffusion, as it operates directly in image space and avoids compression bottlenecks. However, this comes with a significant computational cost due to the high dimensionality of the sampling process.

Latent Diffusion [7] reduces computational complexity by operating in a compressed representation learned by a VAE. While this improves efficiency, it introduces an additional source of information loss: the quality of generated images becomes limited by the expressive power of the latent decoder. This explains the observed reduction in sharpness compared to Pixel Diffusion.
Diffusion models exhibit a clear trade-off between computational efficiency and reconstruction fidelity.

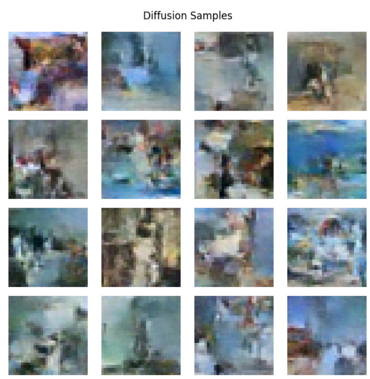

Fig. 8. Pixel Diffusion Samples

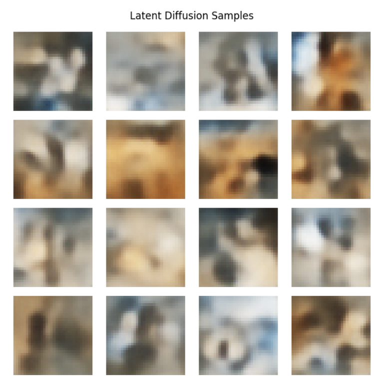

Fig. 9. Latent Diffusion Samples

5.5 Key Insights and Limitations

Overall, the experiments highlight three main findings:

1. Reconstruction quality does not guarantee generative performance (ConvAE vs ConvVAE).
2. Adversarial training is highly effective for sharp image synthesis but unstable.
3. Diffusion models provide a stable but computationally expensive alternative, whose performance depends strongly on scale.

A key limitation of this study is the low image resolution (32×32), which restricts the expressive power of diffusion models and limits the perceptual gap between architectures. Additionally, longer training schedules could further improve diffusion performance.
Despite these constraints, the results clearly illustrate the fundamental trade-offs between reconstruction-based, adversarial, and likelihood-based generative modeling approaches.

5.6 Final Quantitative Evaluation

In this final stage of our analysis, we evaluated the best-performing configurations for each generative family on the full ArtBench-10 dataset. To ensure the statistical robustness of our findings, each evaluation was repeated across 10 random seeds. Table 3 reports the mean and standard deviation for both FID and KID metrics.

Table 3. Final quantitative evaluation on the full ArtBench-10 dataset. Results represent the mean and standard deviation calculated over 10 different random seeds.

Model	FID	KID
ConvVAE	465.25 ± 0.26	0.5643 ± 0.0008
DCGAN	$\mathbf{62.94 \pm 0.50}$	$\mathbf{0.0447 \pm 0.0007}$
Diffusion (Pixel)	82.85 ± 0.44	0.0647 ± 0.0007

DCGAN achieved the best quantitative performance, obtaining the lowest FID and KID values. This indicates that, in the low-resolution ArtBench-10 setting, adversarial training remains highly competitive for matching the visual statistics of the dataset.
Pixel Diffusion ranked second, producing competitive scores and visually consistent samples. Its lower performance relative to DCGAN likely reflects the limited resolution and training budget of the present study.
ConvVAE obtained substantially weaker scores, mainly due to the smoothing effect induced by MSE reconstruction loss and KL regularization, which reduces high-frequency detail.
Overall, the results suggest that model performance on ArtBench-10 is strongly conditioned by image resolution and dataset characteristics. In this constrained low-resolution regime, adversarial training remains highly competitive, while diffusion models may require larger architectures and longer schedules to fully express their advantages.

6 Conclusion

This study compared VAEs, GANs, and Diffusion Models on the ArtBench-10 dataset, revealing clear differences in their generative behavior under a common experimental protocol.
Among the evaluated approaches, adversarial training (DCGAN) proved the most effective in the low-resolution 32×32 setting, consistently generating sharper and more realistic samples. However, this advantage came with increased sensitivity to hyperparameter selection and training instability.
Diffusion models showed strong potential, but their performance was likely constrained by the limited computational budget and image resolution adopted in

this study. These results suggest that diffusion-based methods would benefit substantially from longer training schedules and higher-resolution data.
VAEs remained valuable for learning smooth and interpretable latent spaces, but underperformed in perceptual quality, as reflected by substantially weaker FID and KID scores.
Overall, the experiments highlight a central trade-off in generative modeling: GANs prioritize visual fidelity, VAEs prioritize latent structure, and Diffusion Models offer a flexible compromise that scales effectively with additional compute. Future work should explore hybrid approaches and larger-scale diffusion settings to better combine these complementary strengths.

Declaration of AI Usage

During the preparation of this work, the authors utilized Large Language Models (LLMs) exclusively as formatting and copy-editing assistants. Specifically, AI was employed to assist with LaTeX typesetting, structural organization, and translating rough drafts into formal academic English.
The authors rigorously reviewed, edited, and validated all generated text. The core scientific contributions of this paper—including the conceptualization of the problem, the writing of the Python code, the experimental design, the hyperparameter tuning, the generation of graphs, and the critical analysis of the results—were entirely performed by the human authors. The authors take full responsibility for the scientific integrity, accuracy, and originality of the content presented in this document, and are fully prepared to demonstrate a deep understanding of all technical components during the project defense.

References

1. ArtBench-10 dataset `https://www.kaggle.com/datasets/alexanderliao/artbench10`
2. GenAI_Project: Project Source Code and Data. Arthur Sophiatti and Giovanni Faedo. (2026) `https://github.com/Tuco711/GenAI\_Project`
3. D. P. Kingma and M. Welling, *Auto-Encoding Variational Bayes*, arXiv preprint arXiv:1312.6114, 2013.
4. I. Goodfellow et al., *Generative Adversarial Nets*, Advances in Neural Information Processing Systems (NeurIPS), 2014.
5. A. Radford, L. Metz, S. Chintala, *Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks*, arXiv preprint arXiv:1511.06434, 2015.
6. J. Ho, A. Jain, P. Abbeel, *Denoising Diffusion Probabilistic Models*, Advances in Neural Information Processing Systems (NeurIPS), 2020.
7. R. Rombach et al., *High-Resolution Image Synthesis with Latent Diffusion Models*, Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022.
8. Y. Yu et al., *PixelDiT: Pixel Diffusion Transformers for Image Generation*, arXiv preprint arXiv:2511.20645, 2025.